

CAS CS 565 Notes — Complexity Theory

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1 Syllabus / How the Class Works

- Homeworks, midterm, and final. The midterm is harder than the final.
- Pure theory class. No programming.
- AI is allowed if you acknowledge it.
- Exams are closed book. Handwritten cheat sheets are allowed.
- You can always go back to the professor or TF for regrading. Unless you are right, there is a good chance you lose points.
- If you don't know the answer on a midterm, don't guess. Just say you don't know. You will get 0 points if you guess.
- No makeups for midterms or finals.
- This class will cover deterministic and nondeterministic models.
- This is a grad class.
- Midterms and finals will be on the harder material. Don't expect basics.

2 Deterministic Models, Polynomial Time, and Church's Thesis

- What is computation? What is computable? What is uncomputable?
- Simpler models tend to be better for proofs than complex models, because they are equivalent (in computational power) and easier to reason about.
- View computations as a directed graph $G(V, E)$, where V are events and E are transitions between events.
- If an edge $e \in E$ points from event u to event v , then u is the parent of v , and v is the child of u .
- Each event has a time parameter and a space parameter.
- We will start with rigid models, i.e., models that do not change (are not flexible).
- Defining efficiency:
 - Time complexity: the number of steps it takes to compute the output.
 - Space complexity: the amount of memory it takes to compute the output.
- We will use the following notation:
- There are active and passive computations.
- Sequential computation: when one event is computed at a time.
- Parallel computation: when multiple sequential computations are run in parallel.
- Recall time complexity notation, sup, and inf:
 - Suppose there are two functions, $f : \mathbb{N} \rightarrow \mathbb{R}, n \mapsto f(n)$ and $g : \mathbb{N} \rightarrow \mathbb{R}, n \mapsto g(n)$.
 - $f = O(g) \iff \exists c > 0$ s.t. $f(n) < c \cdot g(n) \forall (n \geq n_0) \iff \sup(\frac{f(n)}{g(n)}) < \infty$
 - $f = \Omega(g) \iff \exists c > 0$ s.t. $f(n) > c \cdot g(n) \forall (n \geq n_0) \iff \inf(\frac{f(n)}{g(n)}) > 0$
 - $f = \Theta(g) \iff f = O(g) \text{ and } f = \Omega(g) \iff \exists c_1, c_2 > 0$ s.t. $c_1 \cdot g(n) < f(n) < c_2 \cdot g(n) \forall (n \geq n_0) \iff \sup(\frac{f(n)}{g(n)}) < \infty \text{ and } \inf(\frac{f(n)}{g(n)}) > 0$
 - $f = o(g) \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$
 - Basically, after some point, n_0 ,
 - * $f = O(g)$ means $f(n) < c \cdot g(n)$
 - * $f = \Omega(g)$ means $f(n) > c \cdot g(n)$
 - * $f = \Theta(g)$ means $c_1 \cdot g(n) < f(n) < c_2 \cdot g(n)$
 - * $f = o(g)$ means $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$
- Some different complexities:
 - $O(1)$: Constant time.
 - $O(\log n) = (\log n)^{O(1)}$: Logarithmic time.

- $O(n) = n^{O(1)}$: Linear time.
- $O(n \log n) = (n \log n)^{O(1)}$: Linearithmic time.
- $O(n^2) = n^{O(2)}$: Quadratic time.
- $O(n^3) = n^{O(3)}$: Cubic time.
- $O(2^n) = 2^{O(n)} = (2^n)^{O(1)}$: Exponential time.
- $O(n!) = n^{O(n)}$: Factorial time.

* Note that:

$$n! = \left(\frac{n}{e}\right)^n \sqrt{2\pi n + t_n}$$

- Typically, $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$ are the ‘best’ time complexities.